The Theory of Everything: D∞ from {⊥, ⊤}

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Abstract

This paper proposes a specific mathematical object as the theory of everything: D∞, the universal reflexive domain constructed from the simplest possible starting domain $D_0 = \{\bot, \top\}$ via Scott's D∞ construction, satisfying $D \cong [D \to D]$.

The proposal follows from a chain of results established across the preceding eleven papers of this research programme. The Self-Referential Relational Principle (SRRP) identifies reality as self-referential relational structure. The mathematical formalisation of self-referential relational structure is $D \cong [D \to D]$. The Infinite Content Theorem proves any reflexive domain has infinite structural content. The Infinite Ground derives that all configurations of reality are realised. D∞ from {⊥, ⊤} is the universal reflexive domain — the one into which all others embed. If reality contains all configurations and has infinite structural content, it must be the most universal solution, which is the one from the simplest seed.

The specific physics we observe — the Standard Model, general relativity, everything measured by experiment — is, in this proposal, what D∞ looks like from a specific position within it. We are nodes in an infinite structure. Our physics is the structure's appearance from our perspective.

The theory is: one distinction ({⊥, ⊤}), one operation (the function space construction), iterated to infinity. Everything emerges.

This proposal is specific, well-defined, and falsifiable. It identifies a concrete mathematical object as reality. The remaining programme is to derive specific known physics from D∞ — to show that gauge groups, particle spectra, spacetime dimensions, and coupling constants emerge from the endomorphism structure of this object. This derivation has not been performed. Until it is, the proposal is the most specific and mathematically well-defined theory of everything candidate available, but it remains a proposal.

Keywords: theory of everything, D∞, reflexive domain, $D \cong [D \to D]$, SRRP, universal domain, {⊥, ⊤}

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1. How We Arrived Here

This section traces the logical chain through eleven papers that leads to the proposal. Each step follows from the previous.

Step 1: The Convergence Finding

The Infinite Ground (Mala, 2026j) introduced the methodology of Convergent Descent and derived that reality is infinite and all configurations are realised. *The Convergence Map* (Mala,

2026o) surveyed five fields and identified five convergences that meta-converge to: reality is self-referential relational structure. *The Convergence Map II* (Mala, 2026p) surveyed nine further fields and arrived independently at the same fixed point through different intermediate convergences. Two surveys, fourteen fields, 140+ established results, one structural characterisation: the SRRP.

Step 2: The Equation

The SRRP Paper (Mala, 2026r) formalised the SRRP mathematically. Self-referential relational structure — a structure whose operations are elements of itself — is precisely what Dana Scott's reflexive domain equation $D \cong [D \to D]$ describes. Scott built it for computation (1972). The programme identified it as the structural equation of reality. This is, to the authors' knowledge, the first equation proposed for the structural nature of reality — describing not how reality behaves but what kind of structure it is.

Step 3: The Mathematical Proofs

Mathematical Foundations (Mala, 2026n) and *Mathematical Foundations II* (Mala, 2026q) proved that reflexive domains necessarily exhibit all nine verified SRRP expressions. The features are a mathematical package deal — you cannot have some without others. The dependency structure forms a web, not a hierarchy. The identity theorem proved that two independently discovered convergence pairs are mathematically identical structures.

Step 4: The Infinite Content

The SRRP Paper proved the Infinite Content Theorem: a reflexive domain contains every structural property it could exhibit — infinitely many — all internally represented as elements of the domain. No internal perspective can see them all. The SRRP has infinite features, not nine. The nine are a verified sample.

Step 5: The Structural Validation

Structural Validation (Mala, 2026s) tested $D \cong [D \to D]$ against four domains: the renormalisation group, quantum observable algebras, classical Hamiltonian mechanics (independent physics test), and game theory (breadth test beyond physics). All four exhibit the structural signature of $D \cong [D \to D]$ with all nine verified SRRP expressions. The equation predicts specific structure. Known physics and mathematics exhibit that structure.

Step 6: Which D?

A natural next question after identifying $D \cong [D \to D]$ as the structural equation: which specific D is reality? One might frame this as a search problem — try different starting domains D_0 until the resulting $D\infty$ reproduces observed physics. But this framing contains an assumption: that the right D is some constrained, specific domain whose endomorphisms happen to match our physics. The present paper questions that assumption and arrives at a different answer.

Step 7: The Present Paper

If reality contains all configurations (Infinite Ground), and a reflexive domain contains all

structural properties (Infinite Content Theorem), then reality must be the MOST universal reflexive domain — not a constrained one. The most universal reflexive domain is $D\infty$ from the simplest seed — the universal domain into which all others embed. And the simplest seed is $\{\bot, \top\}$ — one distinction, the minimum needed for structure to exist at all.

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2. The Theory

2.1 Statement

> **The Theory of Everything.** Reality is $D\infty$ — the universal reflexive domain constructed from $D_0 = \{\bot, \top\}$ via Scott's $D\infty$ construction:
> $D_0 = \{\bot, \top\}$
> $D_{n+1} = [D_n \to D_n]$
> $D\infty = \lim\leftarrow(D_n)$
> satisfying $D\infty \cong [D\infty \to D\infty]$.
> This is a specific, well-defined mathematical object. It is constructed from one distinction and one operation, iterated to infinity. It contains infinite structural content. All physics, mathematics, and computation are endomorphism structure within $D\infty$. The specific physics we observe is $D\infty$ as seen from our position within it.

2.2 Why $\{\bot, \top\}$

The choice of $D_0 = \{\bot, \top\}$ is not arbitrary. It is forced by three converging requirements:

Minimality. $\{\bot, \top\}$ is the simplest non-trivial domain — two elements, one relation. Any simpler (a single point) produces a trivial $D\infty$ with no structure. $\{\bot, \top\}$ is the minimum needed for structure to exist.

Universality. $D\infty$ from $\{\bot, \top\}$ is the universal domain — Scott proved that every countably-based domain is a retract of $D\infty$. "Universal" means every other reflexive domain embeds into it. It is the most encompassing — it contains, in a precise categorical sense, every other countably-based domain as a retract.

Completeness. The Infinite Ground derived that all configurations are realised. The Infinite Content Theorem proved that $D\infty$ has infinite structural content. A more constrained D_0 would produce a $D\infty$ that is a restriction of the universal one — a subset of reality, not reality itself. If reality contains everything, the starting domain must be the one that produces everything.

These three requirements — simplest possible, most universal, contains everything — all point to the same object: $D\infty$ from $\{\bot, \top\}$.

2.3 Why Not a More Complex D_0

One might ask: why not start from a richer D_0 — perhaps one with $SU(3)\times SU(2)\times U(1)$ built in, or one with 3+1 dimensions?

Because putting structure into D_0 means assuming the answer. The power of $D\infty$ from $\{\bot, \top\}$ is that NOTHING is assumed except the bare minimum: one distinction, one operation. If the

Standard Model emerges from this, it emerges necessarily — not because it was built in but because it is a structural consequence of self-referential relational structure iterated to infinity. If it doesn't emerge, the theory is wrong.

This is the strongest possible form of a theory of everything: derive everything from nothing but the simplest possible starting point and one structural principle.

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3. Where Physics Comes From

3.1 The Perspectival Principle

D∞ has infinite structural content. No internal node can see all of it (Structural Inexhaustibility). Every node within D∞ has a finite, partial perspective on the infinite whole. This is SRRP Expression (v) — perspectival irreversibility.

Our physics — the Standard Model, general relativity, the specific constants and symmetries we measure — is what D∞ looks like from our node. Different nodes may see different effective physics, depending on the local structure of D∞ around them. Our node sees SU(3)×SU(2)×U(1) because that is the symmetry structure of D∞ as viewed from where we are within it.

3.2 What Emerges from Iteration

Even the first three levels of explicit computation reveal structural emergence:

Level 0: Two elements. One bit. The simplest distinction.

Level 1: Three elements. Still a chain. One bit has become one trit — a third option (the identity function) has appeared between "all no" and "all yes."

Level 2: Ten elements. The structure BRANCHES for the first time. Elements appear that are incomparable — neither above nor below each other. Multiple directions that cannot be reduced to a single ordering. Something like incompatible observables — measurements that cannot be simultaneously ranked — emerges from nothing but iteration.

Level 3 and beyond: Hundreds, then thousands, then astronomical numbers of elements. Richer partial order structure, potentially non-trivial automorphism groups, increasing width (number of incomparable elements).

The transition from chain to branching at Level 2 is the first qualitative transition — the first emergence of genuine structural complexity from trivial starting conditions. Higher levels produce further qualitative transitions that have not yet been computed.

3.3 What Must Be Shown

For this proposal to be confirmed, specific physics must be derived from D∞:

Gauge structure. The automorphism group of D∞, or a natural subgroup emerging at some level, must contain or produce SU(3)×SU(2)×U(1).

Dimensionality. The structure of D∞ must produce, at some level of description, a notion of dimension — and specifically, 3+1 dimensions must emerge as the natural dimensionality for our perspective.

Particle spectrum. The irreducible representations of the relevant symmetry groups within D∞ must correspond to observed particles.

Coupling constants. Structural parameters of D∞ — ratios, branching factors, spectral properties — must match observed coupling constants, or explain why they take their observed values.

Quantum mechanics and gravity. The formalism of quantum mechanics and the geometry of general relativity must emerge as effective descriptions of D∞ from an internal perspective.

None of this has been derived. Each item is a well-defined mathematical question about a specific mathematical object. The object is known. The questions are stated. The answers are computable, at least in principle.

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4. What This Theory Says

4.1 Reality Is Made of One Thing

Not matter. Not energy. Not spacetime. Not information. One distinction — the difference between ⊥ and ⊤, between less and more, between nothing and something — reflected upon itself infinitely. The function space construction [D → D] is the reflection: the structure's operations becoming elements of itself. Iterated to infinity, this single operation produces everything.

4.2 The Universe Has the Structure of Computation

D ≅ [D → D] is the equation of computation (Scott, 1972) and the equation of reality (this programme). These are the same equation because reality and computation have the same structure — not because reality runs ON a computer, but because both ARE the same kind of self-referential relational structure.

The SRRP takes no stance on whether reality is a simulation. D∞ is compatible with both possibilities. What it says is: whatever reality is — simulation or not — it has the structure of D∞.

4.3 Physics Is Perspectival

There is no "view from nowhere." Every observation is from a position within D∞. The laws of physics are not imposed from outside — they are the structural regularities visible from nodes like ours. Different nodes, different regularities, different "physics." Our physics is not arbitrary — it is constrained by the structure of D∞ — but it is one perspective among infinitely many.

4.4 The Theory Is the Simplest Possible

One distinction. One operation. Iterate. That's it. No free parameters beyond the minimal structural requirement of one distinction. No arbitrary choices. No compactification manifolds,

no string vacua, no landscape of possibilities. The starting point is forced by minimality — the simplest non-trivial domain — and the construction is unique — Scott's D∞. There are no choices to make.

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5. Falsifiability

This theory is falsifiable in the following sense:

D∞ from {⊥, ⊤} is a specific mathematical object with computable properties. If its automorphism structure can be shown to be incompatible with SU(3)×SU(2)×U(1) — if no subgroup or quotient of Aut(D∞) produces the Standard Model gauge group — the theory is wrong. If its structure cannot produce 3+1 dimensional effective descriptions from any internal perspective, the theory is wrong. If specific predictions derivable from D∞ contradict observation, the theory is wrong.

The theory makes a specific, checkable claim: reality is THIS mathematical object. The object is well-defined. Its properties are in principle computable. If the properties don't match observation, the theory fails.

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6. Novel Contributions

1. A specific mathematical object proposed as the theory of everything. D∞ from D_0 = {⊥, ⊤}. Not a framework, not a programme, not a class of models — a single, well-defined mathematical object identified as reality itself. This has not been previously proposed.

2. The logical derivation from the SRRP. The step-by-step chain from convergence evidence through mathematical formalisation through infinite content through universality to the specific identification of D∞ from {⊥, ⊤}. Each step follows from the previous. The chain is presented here for the first time.

3. The perspectival account of specific physics. The proposal that the specific physics we observe (Standard Model, general relativity, coupling constants) is what D∞ looks like from our position within it — that physics is perspectival, not absolute, but constrained by D∞'s structure. This specific account has not been previously proposed.

4. The simplest possible theory of everything. One distinction, one operation, no free parameters beyond the minimal structural requirement. The starting domain is forced by minimality. The construction is unique. There are no choices to make.

5. Explicit identification of what remains to be shown. The specific mathematical questions — gauge structure, dimensionality, particle spectrum, coupling constants, quantum mechanics, gravity — that must be answered about D∞ for the theory to be confirmed.

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7. Conclusion

The theory of everything, as proposed here, is:

$D_0 = \{\bot, \top\}$

$D_{n+1} = [D_n \to D_n]$

$D\infty = \lim\leftarrow(D_n)$

$D\infty \cong [D\infty \to D\infty]$

One distinction. One operation. Iterated to infinity. Everything emerges.

This is a specific, falsifiable proposal with no free parameters beyond the minimal structural requirement of one distinction. It identifies a concrete mathematical object — well-defined, computable, unique — as reality itself. It is derived from the convergence of established results across fourteen fields through a chain of logical steps, each building on the last. It is the simplest possible theory: the minimum starting point, the minimum operation, the minimum assumptions.

Whether it is correct depends on whether specific physics can be derived from D∞. That derivation is the remaining programme — the most important open problem this work identifies. The object is known. The question is whether it is our universe.

If it is, then everything — every particle, every force, every dimension, every law — emerges from one bit and one operation. The theory of everything is the theory of the simplest possible thing, iterated without limit.

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